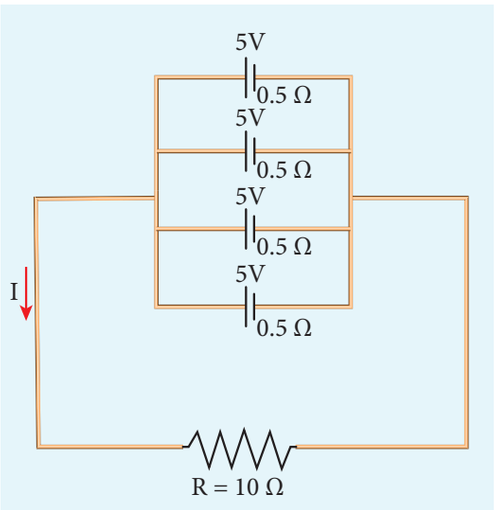
The above equation implies that current due to the whole battery is the same as that due to a single cell. Hence it is advantageous to connect cells in parallel when the external resistance is very small compared to the internal resistance of the cells.



When the car engine is started with headlights turned on, they sometimes become dim. This is due to the internal resistance of the car battery.

EXAMPLE 2.19

For the given circuit



Find

i) Equivalent emf
ii) Equivalent internal resistance
iii) Total current (I)
iv) Potential difference across each cell
v) Current from each cell

Solution

i) Equivalent emf $\varepsilon_{eq} = 5$ V

ii) Equivalent internal resistance,

$$R_{eq} = \frac{r}{n} = \frac{0.5}{4} = 0.125\Omega$$

iii) total current, $I = \dfrac{\varepsilon}{R + r/n}$

$$I = \frac{5}{10 + 0.125} = \frac{5}{10.125}$$

$$I \approx 0.5 \text{ A}$$

iv) Potential difference across each cell

$$V = IR = 0.5 \times 10 = 5 \text{ V}$$

v) Current from each cell, $I' = \dfrac{I}{n}$

$$I' = \frac{0.5}{4} = 0.125 \text{ A}$$

2.5 KIRCHHOFF'S RULES

Ohm's law is useful only for simple circuits. For more complex circuits, Kirchhoff's rules can be used to find current and voltage. There are two generalized rules: i) Kirchhoff's current rule ii) Kirchhoff's voltage rule.

2.5.1 Kirchhoff's first rule (Current rule or Junction rule)

It states that the algebraic sum of the currents at any junction of a circuit is zero. It is a statement of law of conservation of electric charge. The charges that enter a given junction in a circuit must leave that junction since charge cannot build up or disappear at a junction. By convention, current entering the junction is taken as positive and current leaving the junction is taken as negative.

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